Logistic Regression: A Bernoulli Whose Parameter Is a Function of X

Yuval Lavie

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Abstract

Binary classification, seen from statistics: the label given the features is *always* a Bernoulli variable — the only modeling decision is the form of its parameter function $p(x) = \Pr(Y{=}1 \mid X{=}x)$. The linear classifier bets on linear log-odds, $p(x) = \sigma(w^\top x)$, and conditional maximum likelihood then reproduces, composed, the two gradients of the previous notes. New phenomena: the stationarity equations are transcendental — the first model in this collection with *no closed form* — yet the loss is convex, Newton's method (IRLS) solves it in a handful of weighted least-squares steps, the fitted classifier approaches the Bayes floor, and on separable data the MLE fails to exist. All demonstrated in the companion scripts, one solver per file over a shared `common.py`: `logistic_sgd.py` (the per-sample loop, logged, with the autograd identity check), `logistic_gd.py` (full gradient; geometric decay of $\|\nabla\|$ in the log; the separable-data divergence), `logistic_newton.py` (Newton/IRLS; watch $\|\nabla\|$ square away in eight rows), and `logistic_animation.py` (the SGD film). Each ends with a held-out classification report.

1 The conditional-Bernoulli view

Let (X, Y) be jointly distributed with $Y \in \{0, 1\}$. Whatever the joint $\mathcal{D}$ is, it factors as a marginal over x times a conditional over y — and a distribution on $\{0, 1\}$ *is* a Bernoulli, so with no assumption at all:

$$Y \mid X = x \;\sim\; \text{Bernoulli}\big(p(x)\big), \qquad p(x) \;=\; \Pr(Y = 1 \mid X = x) \;=\; \mathbb{E}[Y \mid X = x]. \tag{1}$$

The object of interest is not one random variable but the *family* of Bernoullis indexed by x — equivalently the single function $p : \mathcal{X} \to [0, 1]$ (the *regression function*). "$Z = Y|X$ is Bernoulli with parameter depending on X" is exactly this, stated for each slice $X = x$.

Where the assumption actually lives. Bernoulli-ness in (1) is forced by the binary support and costs nothing. The entire model is the choice of a *form* for $p(x)$. The constant choice $p(x) \equiv p$ is the `bernoulli/` experiment; the linear classifier's choice cannot be $p(x) = w^\top x$ (a line escapes $[0, 1]$), so linearity is placed in the *log-odds*:

$$\log \frac{p(x)}{1 - p(x)} \;=\; w^\top x \qquad \Longleftrightarrow \qquad p(x) \;=\; \sigma(w^\top x) \;=\; \frac{1}{1 + e^{-w^\top x}}, \tag{2}$$

with x carrying a fixed leading 1 so w includes the intercept. In GLM language: Bernoulli response, logit link. The decision boundary $\{x : p(x) = \frac{1}{2}\}$ is the hyperplane $w^\top x = 0$ — the "linear" of the linear classifier.

2 Conditional maximum likelihood

Given samples $(x_i, y_i)_{i=1}^n$, condition on the observed x_i and apply (1): the conditional likelihood and (after the usual log, $\div n$, negate — each optimizer-preserving) the average NLL are

$$L(w) = \prod_{i=1}^n p(x_i)^{y_i}\big(1-p(x_i)\big)^{1-y_i}, \qquad \mathcal{L}(w) = -\frac{1}{n}\sum_{i=1}^n \Big[y_i \log p(x_i) + (1-y_i)\log\big(1-p(x_i)\big)\Big], \quad (3)$$

with $p(x_i) = \sigma(w^\top x_i)$ — the *cross-entropy loss*, which by the information-theory note is $H(\text{data}) + \mathrm{KL}(\text{data}\,\|\,\text{model})$ in conditional form.

The gradient composes the previous two notes. Per sample, write $z_i = w^\top x_i$. The Bernoulli note gives $\partial\mathcal{L}_i/\partial z_i = \sigma(z_i) - y_i$ (the chain rule through $\sigma' = \sigma(1-\sigma)$, everything cancelling); the linear-regression note gives $\partial z_i/\partial w = x_i$. Chained:

$$\nabla_w \mathcal{L}_i \;=\; \big(\sigma(w^\top x_i) - y_i\big)\, x_i, \qquad \nabla_w \mathcal{L} \;=\; \frac{1}{n} X^\top\big(\sigma(Xw) - y\big) \quad (4)$$

— *Bernoulli residual times feature vector*: the same "prediction minus target, pushed along the features" shape as least squares, with the sigmoid interposed.

3 The closed form dies — and what survives

Setting (4) to zero:

$$\sum_{i=1}^n \sigma(w^\top x_i)\, x_i \;=\; \sum_{i=1}^n y_i\, x_i. \quad (5)$$

Sigmoids of the unknowns, summed: *transcendental* in w. Bernoulli gave $\hat{p} = m/n$; least squares gave the normal equations; here algebra produces nothing — the first model in this collection where the theoretical route genuinely does not exist. Not for lack of good structure:

Proposition 1. *$\mathcal{L}$ is convex: its Hessian is*

$$\nabla^2 \mathcal{L}(w) \;=\; \frac{1}{n} X^\top S X \;\succeq\; 0, \qquad S = \mathrm{diag}\big(p_i(1-p_i)\big), \quad (6)$$

the weights being exactly the conditional Bernoulli variances. Hence every stationary point is a global minimum (optimization note), and gradient methods inherit one basin.

Newton / IRLS: the statistician's solver. Convex + smooth + cheap Hessian invites Newton's method (optimizers note): $w \leftarrow w - (X^\top S X)^{-1} X^\top (p - y)$ (the $1/n$ cancels). Rearranged, each iteration solves the *weighted least-squares* system $X^\top S X\, d = X^\top(y - p)$ — *iteratively reweighted least squares*: logistic regression as a sequence of linear regressions, reweighted by the current Bernoulli variances, with quadratic convergence (the script needs 8 steps to exhaust float64 while GD takes thousands). This is the classical fit route in statistics packages.

Remark 1 (When the MLE does not exist: separable data)**.** *If some w classifies the sample perfectly ($w^\top x_i > 0$ exactly when $y_i = 1$), then scaling $w \to c\,w$ with $c \to \infty$ drives every $p(x_i)$ to its label and $\mathcal{L} \to 0$ without attaining it: the infimum is not achieved, $\|w\|$ diverges, and probabilities saturate. The script watches $\|w\|$ grow without bound under GD on separable data. Cures: ℓ_2 regularization (adds $\frac{\lambda}{2}\|w\|^2$, restoring strong convexity and a finite optimum) or early stopping. Ironically, "too easy" data breaks the estimator.*

4 The Bayes floor

Whatever classifier h anyone builds from x, its error obeys

$$\Pr\big(h(X) \neq Y\big) \;\geq\; \mathbb{E}\big[\min\{p(X),\, 1 - p(X)\}\big], \tag{7}$$

with equality for the *Bayes classifier* $h^*(x) = \mathbf{1}[p(x) > \frac{1}{2}]$: at each x you cannot beat predicting the likelier label, and its mistake probability is the smaller of the two. The floor is a property of the *distribution* (how often $p(x)$ is near $\frac{1}{2}$), not of any algorithm — it is the label noise of the learnability note, seen from the statistics side. Because our experiment is well-specified (data genuinely logistic), conditional MLE is consistent for w^*, and the fitted classifier's error approaches (7).

Remark 2 (Misspecification)**.** *If the true $p(x)$ is* not *of the form* (2)*, the MLE does not break — by the information-theory note it converges to the KL projection of the truth onto the logistic family: the best logistic approximation, in exactly the divergence that cross-entropy measures. Well-specified consistency and misspecified KL projection are the same theorem seen twice.*

5 Experiment

$n = 5000$, $x \sim \mathcal{N}(0, I_2)$, $w^* = (0.5,\, 2,\, -1.5)$, labels drawn from (1) (seed 7). Newton/IRLS and GD agree on the unique optimum ($\hat{w} \approx w^*$ to sampling accuracy); hand and autograd SGD trajectories coincide to 10^{-10}; the fitted classifier's test error matches the Bayes floor $\mathbb{E}[\min(p, 1-p)]$ to three decimals; and the third panel plots fitted $\sigma(\hat{w}^\top x)$ against true $p(x)$ on fresh points — a diagonal: what was estimated is not a label rule but *the Bernoulli parameter as a function of* x, which is where this note began.